Exercise 3

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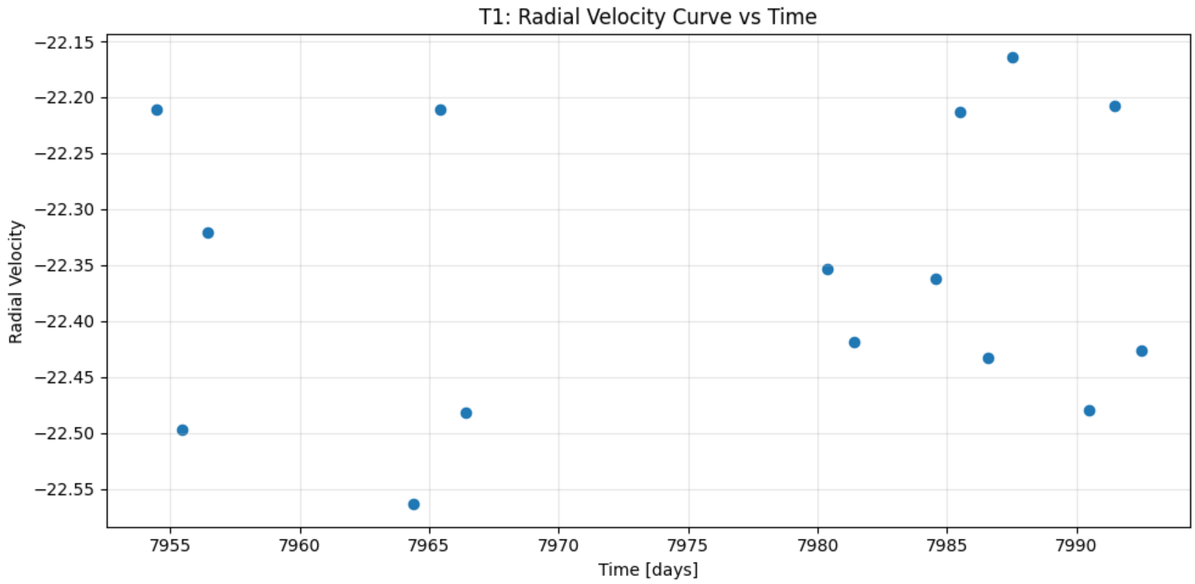
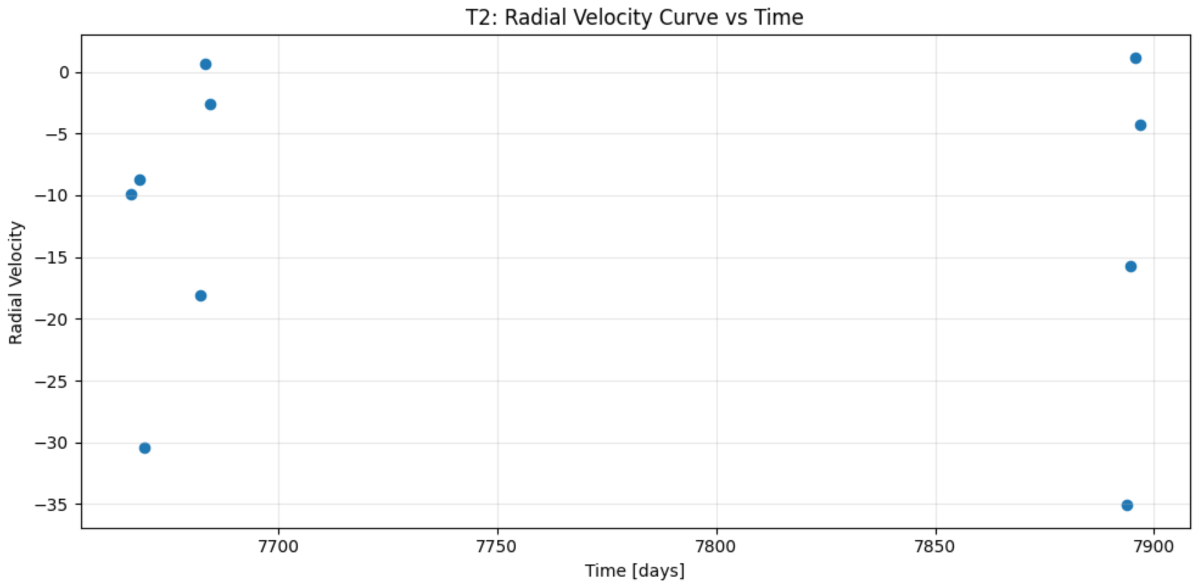
3.1





The radial velocity data were plotted first as a function of time and then phase-folded using the previously determined transit ephemeris. The phase was calculated from the orbital period P and mid-transit epoch E. In the phase-folded RV curve, the periodic stellar motion becomes clear because measurements from different orbital cycles are placed into one common orbital cycle.

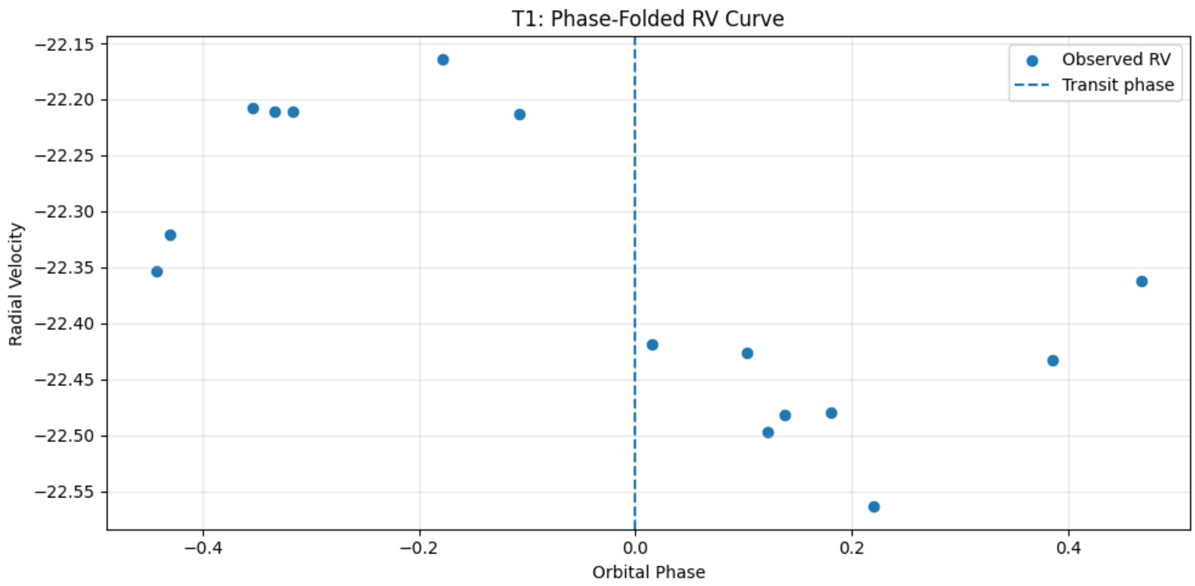
3.2:

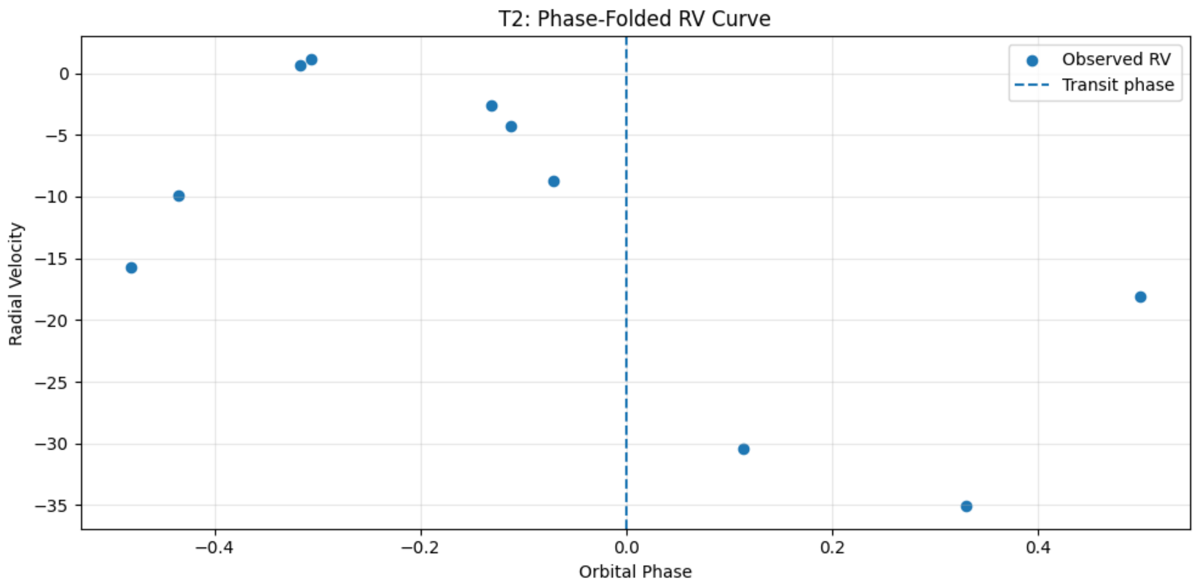
The phase-folded radial velocity curves show a constant vertical offset, corresponding to the systemic radial velocity v0 of the observed star-planet system. This offset represents the average line-of-sight velocity of the entire system relative to the observer and is unrelated to the orbital reflex motion itself. Physically, v0 originates from the bulk motion of the star through the Galaxy, combined with the observer's reference frame. The orbital motion produced by the planet is therefore superimposed on top of this constant background velocity.

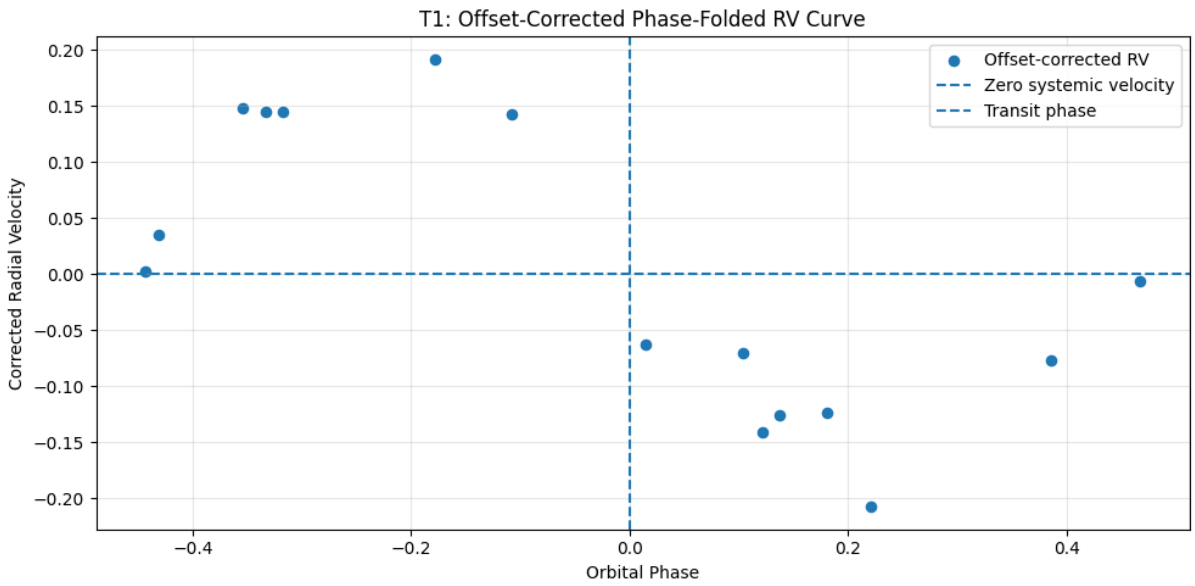
For T1: v0 = -22.356 ± 0.009 km s^{-1}

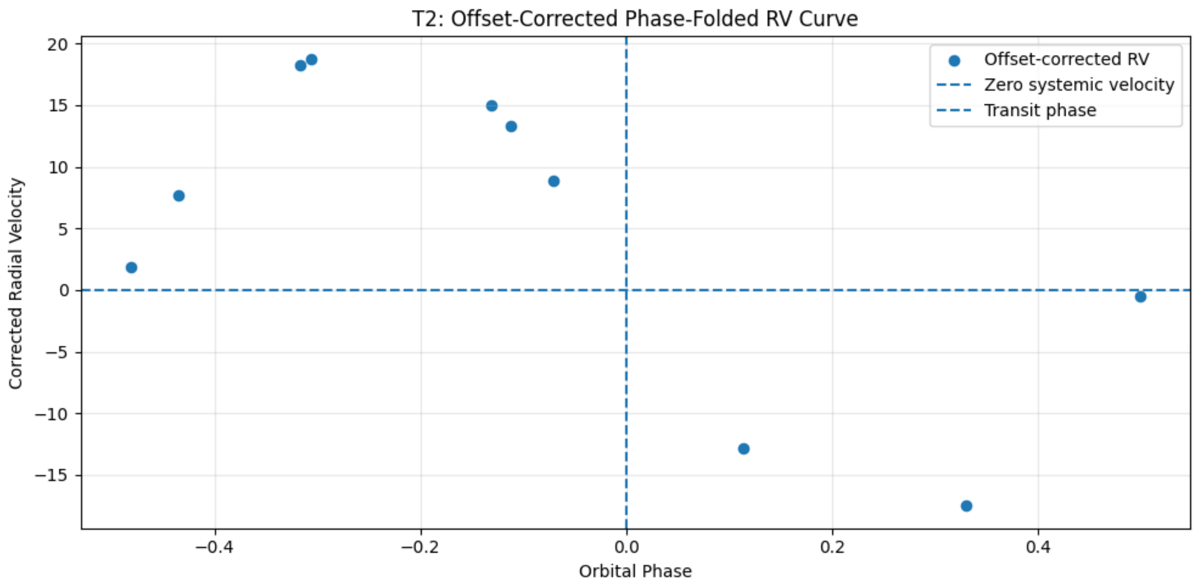
For T2: v0 = -17.590 ± 0.111 km s^{-1}

After subtracting these offsets, the corrected RV curves oscillate around zero velocity. The planetary transit occurs at phase 0 in the phase-folded RV curve.









3.3:

The phase-folded radial velocity curves were fitted using:

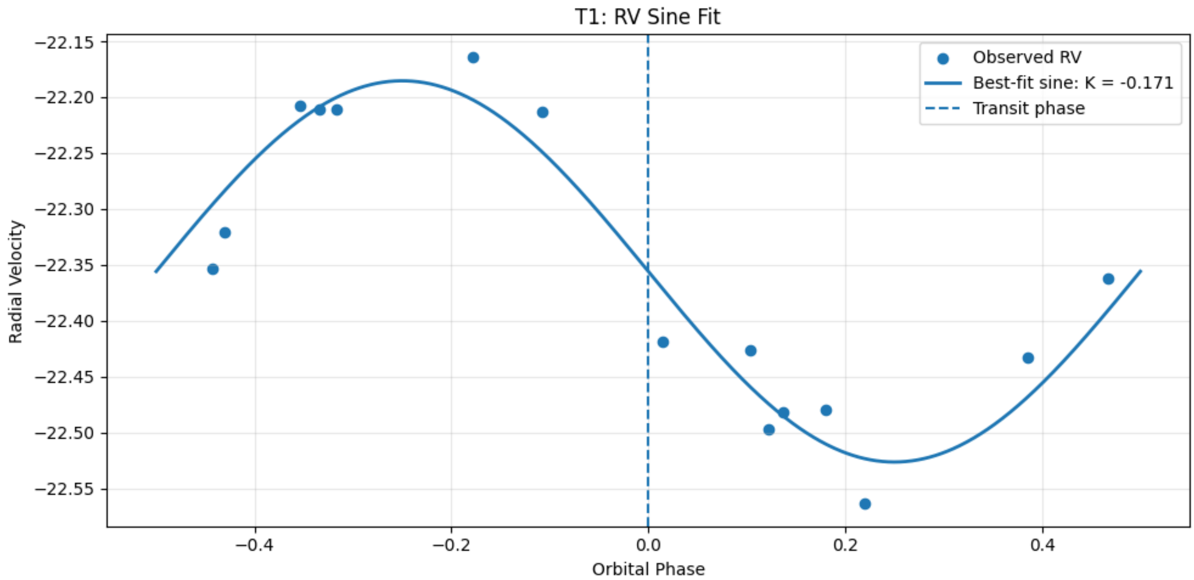
vr = K sin(2πφ) + v0

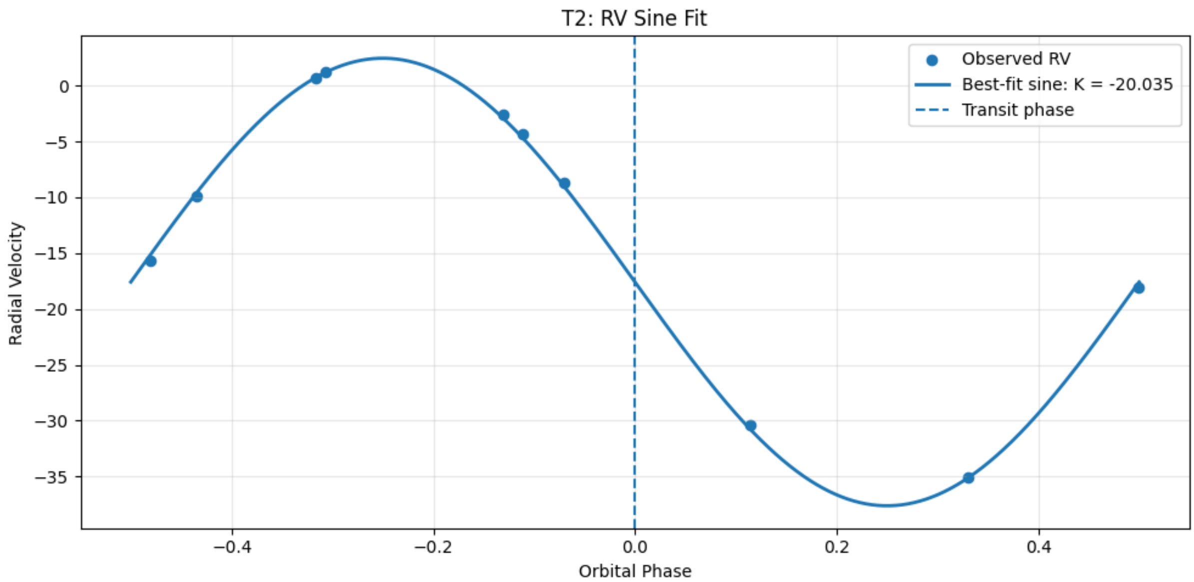
where: K = radial velocity semi-amplitude, φ = orbital phase, v0 = systemic velocity offset

For T1: K = 0.171 ± 0.013 km s^{-1}

For T2: K = 20.035 ± 0.171 km s^{-1}

The RV semi-amplitude K measures the strength of the stellar reflex motion induced by the orbiting companion. The significantly larger K value for T2 indicates a substantially stronger gravitational influence on the host star compared to T1.





Final Fitted Values

T1: v0 = -22.356 ± 0.009 km s^{-1}, K = 0.171 ± 0.013 km s^{-1}

T2: v0 = -17.590 ± 0.111 km s^{-1}, K = 20.035 ± 0.171 km s^{-1}